

The 10X Engineer

A semester of engineering practice with AI agents

Not a replacement for the engineer, a tool the engineer manages. The course teaches you to work as a first line manager of AI agents: define a task, break it down, give the agent the right context, check the result and take responsibility for it.

Most of the work happens at the keyboard. Each topic opens with a short lecture and continues as practice. Alongside the practice we cover the theory, from classical machine learning to transformers, so that working with the tools rests on an understanding of what happens inside them. At the end of the semester each participant presents a working project in their own professional field.

13 sessions

- 1 Introduction to the 10X engineer: three levels of working with AI
- 2 The user level: what a chat window can do, and where it stops
- 3 Skills, APIs, the MCP protocol and project memory
- 4 Formal process, specifications and the virtual AI team
- 5 ESP32 hackathon: two hours, tasks revealed on the day **competition**
- 6 Theory 1: classical AI, from statistics to the eve of the transformer
- 7 Theory 2: large language models and transformers **train a model**
- 8 Inside the harness: how an IDE runs an agent **build one**
- 9 The 10X engineer on the exponential slope
- 10 Physical AI: systems that sense and act, and research at the Technion
- 11 Personal project: definition and start of work
- 12 Project presentations, two sessions **finals**

Course details

Day and time

Mondays, 08:30

Room

430

Length

13 sessions

Open to

Undergraduate and graduate students

Places

Limited

What to bring

- A personal laptop for the whole semester
- A paid AI coding assistant, on the order of 20 dollars per month
- An IDE, for example Visual Studio Code or Cursor

Instructors



Avi Salmon

Innovation lead at Intel Israel and chip design engineer



Yuval Vered

Leads an AI chip development group at Intel



Register through the faculty office

Scan the code for the full syllabus, session topics and grading

avisalmon.github.io/10XengineerSyllabus